SET 29: STUDENT ACADEMIC PERFORMANCE ANALYSIS

Dataset: Student Academic Performance Data

Department	AIDS
Course Code / Course Name	DSA0602 / Data Handling and Visualization
Experiment	Data Visualization using R and Tableau
Name	K. Deeksha Sree
Register Number	192424097
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Day	Thursday

Student_ID	Age	Study_Hours	Attendance (%)	Test_Score	Participation_Score
S1	19	12	90	85	8
S2	21	8	70	70	7
S3	20	15	95	92	9
S4	22	10	85	80	8
S5	23	7	60	65	6

PROGRAM 1: Stacked Area Chart of Test Score and Participation Score

Question: Using R, create a stacked area chart showing Test_Score and Participation_Score across students. Analyze trends.

```
# Dataset
students <- data.frame(
  Student_ID          = c("S1", "S2", "S3", "S4", "S5"),
  Age                 = c(19, 21, 20, 22, 23),
  Study_Hours         = c(12, 8, 15, 10, 7),
  Attendance          = c(90, 70, 95, 85, 60),
  Test_Score          = c(85, 70, 92, 80, 65),
  Participation_Score = c(8, 7, 9, 8, 6)
)

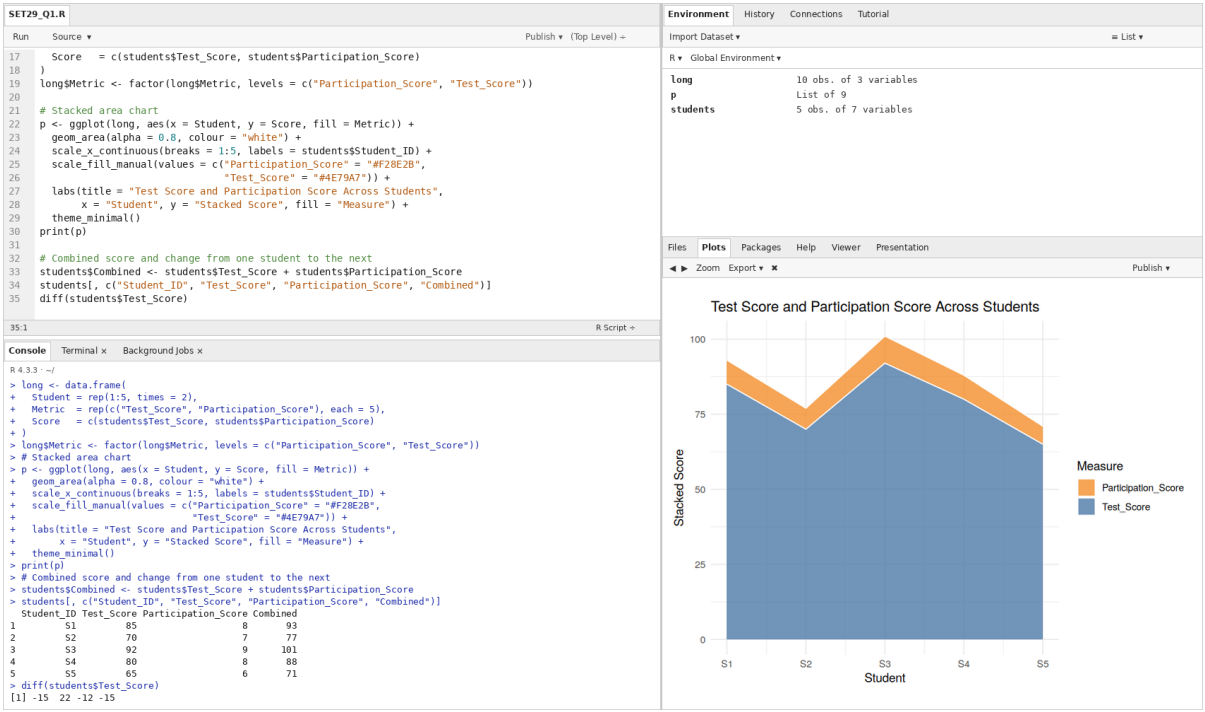
library(ggplot2)

# Long format for stacking
long <- data.frame(
  Student = rep(1:5, times = 2),
  Metric  = rep(c("Test_Score", "Participation_Score"), each = 5),
  Score   = c(students$Test_Score, students$Participation_Score)
)
long$Metric <- factor(long$Metric, levels = c("Participation_Score", "Test_Score"))

# Stacked area chart
p <- ggplot(long, aes(x = Student, y = Score, fill = Metric)) +
  geom_area(alpha = 0.8, colour = "white") +
  scale_x_continuous(breaks = 1:5, labels = students$Student_ID) +
  scale_fill_manual(values = c("Participation_Score" = "#F28E2B",
                               "Test_Score" = "#4E79A7")) +
  labs(title = "Test Score and Participation Score Across Students",
       x = "Student", y = "Stacked Score", fill = "Measure") +
  theme_minimal()
print(p)

# Combined score and change from one student to the next
students$Combined <- students$Test_Score + students$Participation_Score
students[, c("Student_ID", "Test_Score", "Participation_Score", "Combined")]
diff(students$Test_Score)
```

Output 1: Stacked Area Chart



Observation: Test scores go 85, 70, 92, 80, 65 across S1 to S5, so they change by -15, +22, -12 and -15 between neighbouring students. The combined score peaks at S3 (101) and is lowest at S5 (71). Participation scores (8, 7, 9, 8, 6) rise and fall in the same places as test scores (correlation 0.976), so the more engaged students also scored higher. Because participation is on a 1 to 10 scale and test score on a 100 scale, the orange participation band is thin and the stacked total is dominated by Test_Score. The student order is only an ID sequence, so there is no steady upward or downward trend over time: the pattern is an alternating zig-zag with a decline from S3 to S5.

PROGRAM 2: Boxplot of Study Hours by Attendance Quartile

Question: Using R, generate a boxplot of Study_Hours grouped by Attendance quartiles. Interpret patterns. Include axis labels, title, and colors for each quartile. Analyze the boxplot to

identify median and interquartile range of study hours per attendance quartile and any outliers in study hours.

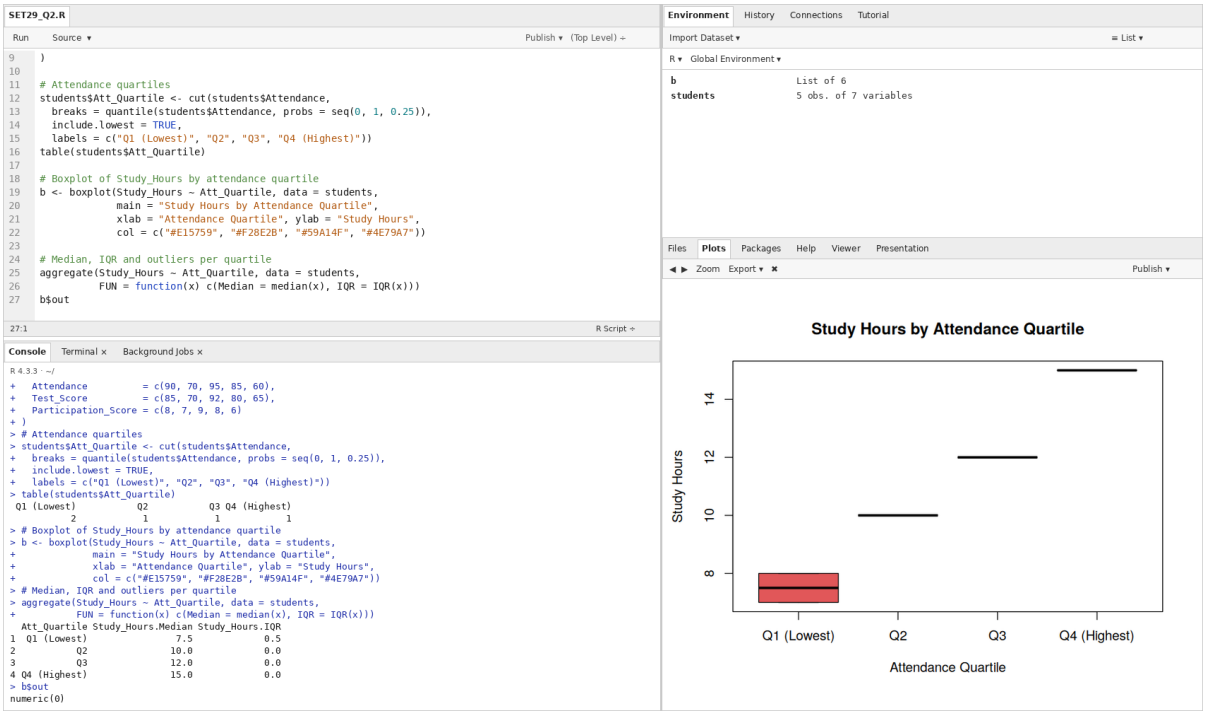
```
# Dataset
students <- data.frame(
  Student_ID          = c("S1", "S2", "S3", "S4", "S5"),
  Age                 = c(19, 21, 20, 22, 23),
  Study_Hours         = c(12, 8, 15, 10, 7),
  Attendance          = c(90, 70, 95, 85, 60),
  Test_Score          = c(85, 70, 92, 80, 65),
  Participation_Score = c(8, 7, 9, 8, 6)
)

# Attendance quartiles
students$Att_Quartile <- cut(students$Attendance,
  breaks = quantile(students$Attendance, probs = seq(0, 1, 0.25)),
  include.lowest = TRUE,
  labels = c("Q1 (Lowest)", "Q2", "Q3", "Q4 (Highest)"))
table(students$Att_Quartile)

# Boxplot of Study_Hours by attendance quartile
b <- boxplot(Study_Hours ~ Att_Quartile, data = students,
             main = "Study Hours by Attendance Quartile",
             xlab = "Attendance Quartile", ylab = "Study Hours",
             col = c("#E15759", "#F28E2B", "#59A14F", "#4E79A7"))

# Median, IQR and outliers per quartile
aggregate(Study_Hours ~ Att_Quartile, data = students,
          FUN = function(x) c(Median = median(x), IQR = IQR(x)))
b$out
```

Output 2: Boxplot of Study Hours by Attendance Quartile



Observation: The quartile cut-points for attendance are 60, 70, 85, 90 and 95. Q1 (lowest) holds S2 and S5 and has a median of 7.5 study hours with an IQR of 0.5. Q2, Q3 and Q4 each hold one student, so their boxes collapse to a line at 10, 12 and 15 hours (IQR 0). The median study hours therefore rise steadily with attendance (7.5, 10, 12, 15), meaning students who attend more also study more. No outliers were found (b$out is numeric(0)). Since three of the four groups contain a single student, the boxes are degenerate and the pattern should be confirmed with a larger class.

PROGRAM 3: Density Plot of Test Score

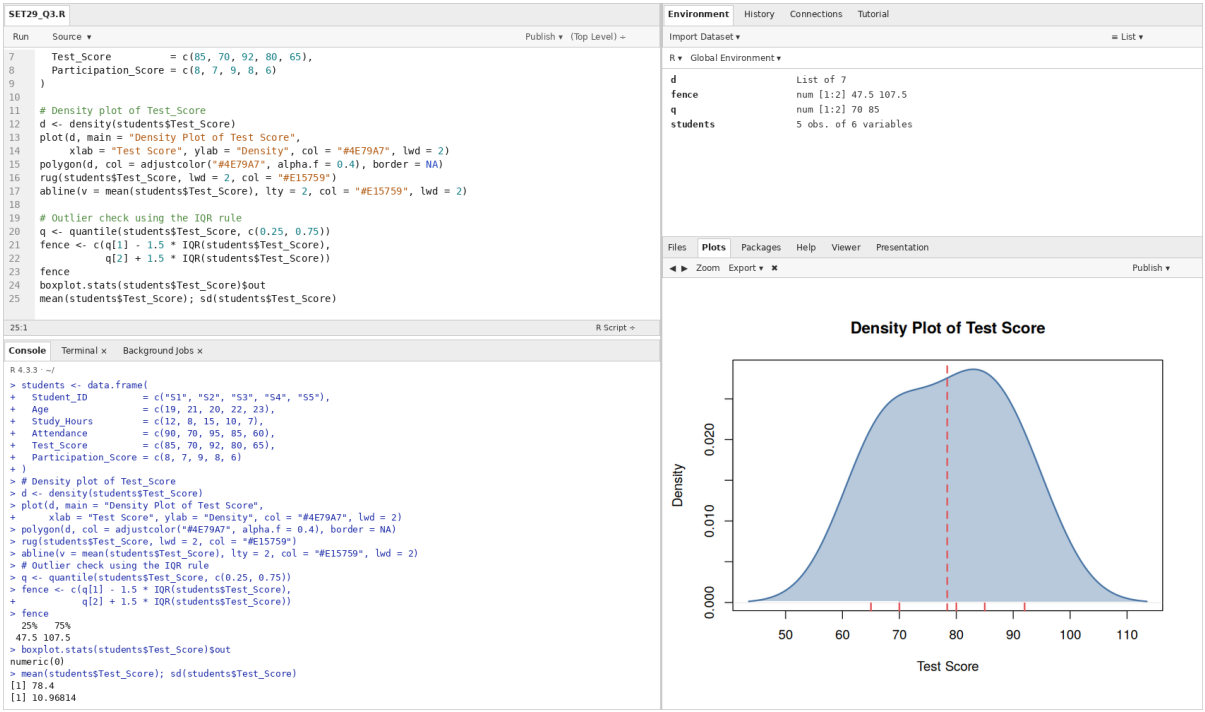
Question: Using R, construct a density plot for Test_Score to evaluate score distribution and outliers.

```
# Dataset
students <- data.frame(
  Student_ID          = c("S1", "S2", "S3", "S4", "S5"),
  Age                 = c(19, 21, 20, 22, 23),
  Study_Hours         = c(12, 8, 15, 10, 7),
  Attendance          = c(90, 70, 95, 85, 60),
  Test_Score          = c(85, 70, 92, 80, 65),
  Participation_Score = c(8, 7, 9, 8, 6)
)

# Density plot of Test_Score
d <- density(students$Test_Score)
plot(d, main = "Density Plot of Test Score",
     xlab = "Test Score", ylab = "Density", col = "#4E79A7", lwd = 2)
polygon(d, col = adjustcolor("#4E79A7", alpha.f = 0.4), border = NA)
rug(students$Test_Score, lwd = 2, col = "#E15759")
abline(v = mean(students$Test_Score), lty = 2, col = "#E15759", lwd = 2)

# Outlier check using the IQR rule
q <- quantile(students$Test_Score, c(0.25, 0.75))
fence <- c(q[1] - 1.5 * IQR(students$Test_Score),
           q[2] + 1.5 * IQR(students$Test_Score))
fence
boxplot.stats(students$Test_Score)$out
mean(students$Test_Score); sd(students$Test_Score)
```

Output 3: Density Plot of Test Score



Observation: The density curve is a single broad hump peaking in the mid-80s, with the mean (78.4, red dashed line) slightly below the median (80), which indicates a mild left skew caused by the lower scores of 65 and 70. The standard deviation is 10.97. The IQR rule gives fences of 47.5 and 107.5, and all scores lie between 65 and 92, so there are no outliers. The curve extends past 100 and below 50 only because kernel smoothing spreads the five points out, not because such scores exist. With five observations the shape of the density is only a rough guide.

PROGRAM 4: Interactive Scatter Plot in Tableau

Question: In Tableau, design an interactive scatterplot with Study_Hours on X-axis, Test_Score on Y-axis, colored by Participation_Score. Include filters for Age.

Tableau Steps

1. Open Tableau and connect to the student data file (Excel/CSV). Confirm that Age, Study_Hours, Attendance, Test_Score and Participation_Score are numeric measures and Student_ID is a dimension.

2. Create a worksheet named "Study vs Test Score". Drag Study_Hours to Columns and Test_Score to Rows.

3. Drag Student_ID to Detail so each student is one point (or turn off Analysis > Aggregate Measures). Choose Circle in the Marks card and increase the size.

4. Drag Participation_Score to Color. Edit the colour to a sequential palette (for example Orange-Red) so higher participation is darker.

5. Drag Student_ID to Label and add Age, Attendance and Participation_Score to the Tooltip so hovering shows full details.

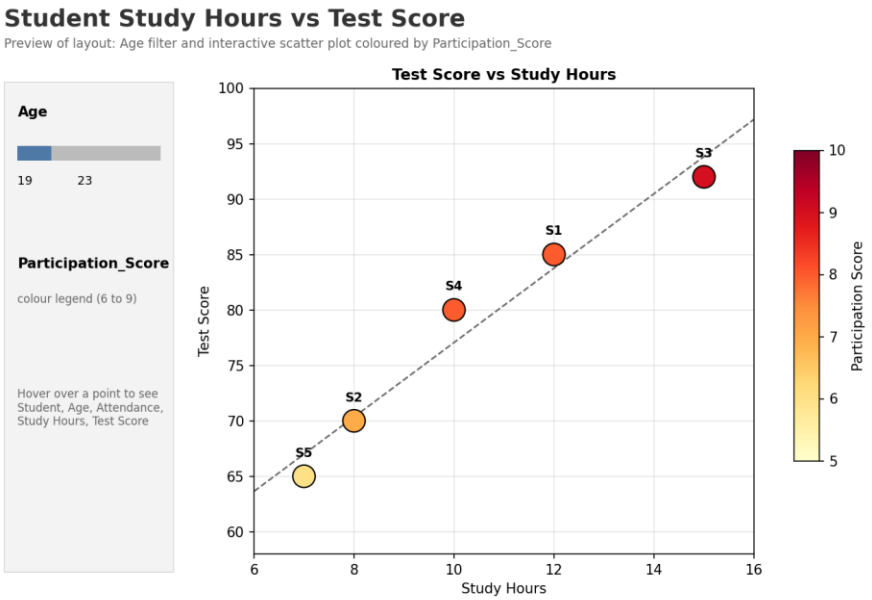
6. Add a trend line from the Analytics pane (Trend Line, Linear) and title the sheet "Test Score vs Study Hours" with axis titles.

7. Add the Age filter: drag Age to Filters, choose Range of values, right-click the filter and select Show Filter. Display it as a slider.

8. Create a New Dashboard, set the size to Automatic, drag in the scatter plot and add the title "Student Study Hours vs Test Score". Show the Age filter and the colour legend.

9. Test the dashboard by moving the Age slider and hovering over points to check the tooltips.

Output 4: Interactive Scatter Plot Dashboard (layout preview)



Observation: The scatter plot shows a strong positive relationship between Study_Hours and Test_Score (r = 0.98). S3 (15 hours, 92 marks, participation 9) is at the top right in the darkest colour, while S5 (7 hours, 65 marks, participation 6) is at the bottom left in the lightest colour, so high participation goes with long study time and high scores. The Age slider (19 to 23) lets the viewer narrow the plot to a group of ages, but the oldest student (S5, age 23) has the lowest score and the youngest (S1, age 19) is second highest, so age alone does not explain performance in this small sample.

Procedure

1. Create the student data frame in R.

2. Reshape Test_Score and Participation_Score to long format and draw the stacked area chart using geom_area().

3. Divide Attendance into quartiles with cut() and quantile(), draw the coloured boxplot of Study_Hours with boxplot(), and compute the median and IQR of each quartile.

4. Draw the density plot of Test_Score with density() and check outliers using the IQR rule and boxplot.stats().

5. Import the data into Tableau and build the Study_Hours vs Test_Score scatter plot coloured by Participation_Score.

6. Add the Age filter, build the dashboard and test the interactivity.

Result

The Student Academic Performance data was successfully visualized using a stacked area chart (participation moves with test score), a boxplot of study hours by attendance quartile (median study hours rise from 7.5 to 15 with attendance, no outliers), a density plot of test scores (mean 78.4, no outliers) and an interactive Tableau scatter plot with an Age filter.